EXPERIMENT #8

Timestamp: 20260317_094649 | Environment: Local

1. Experiment Parameters & Network Statistics

Parameter	Value
Number of sticks (N)	30
Stick length (a)	0.7
Domain size	1.0 × 1.0
Source region	x < 0.1
Drain region	x > 0.9
Total intersections	64
Has percolation	✓ Yes
# Percolating paths	1
Nodes before pruning	63
Nodes after pruning	9
Edges before pruning	62
Edges after pruning	8

2. Network Visualizations

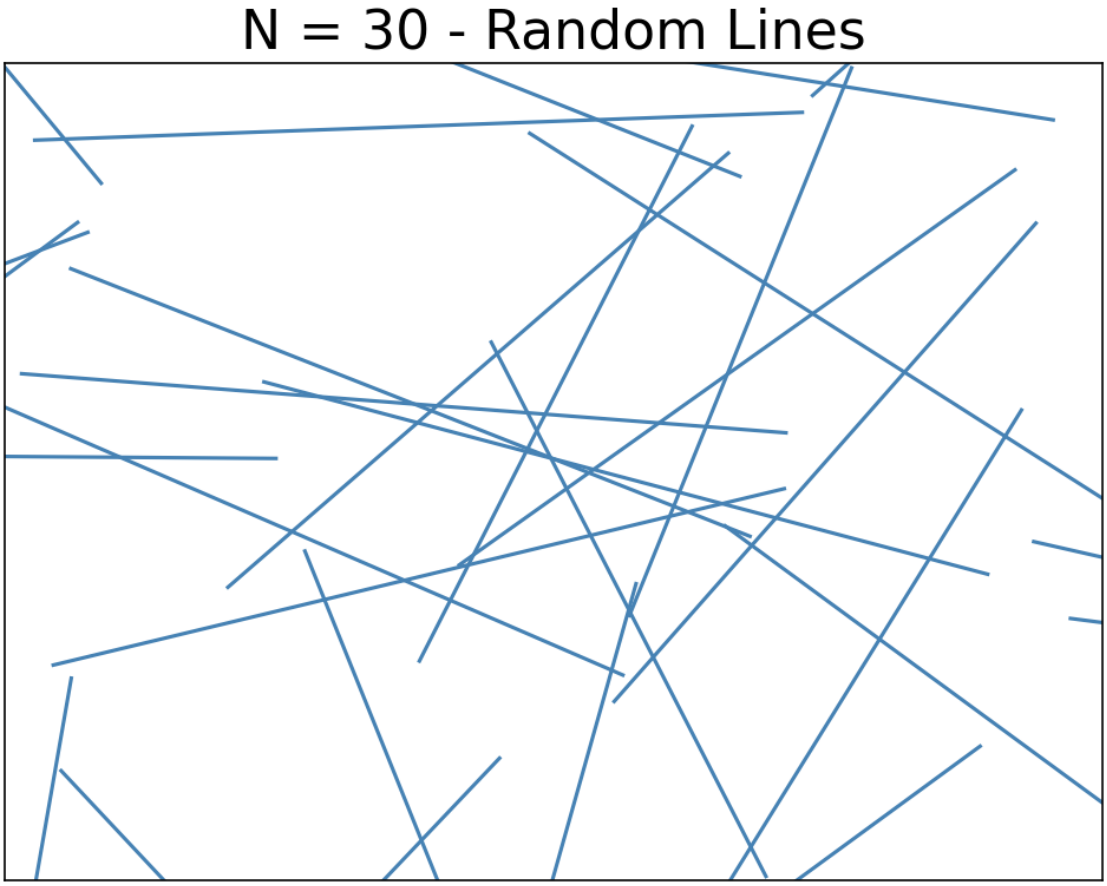


Fig. — 01 random lines N30

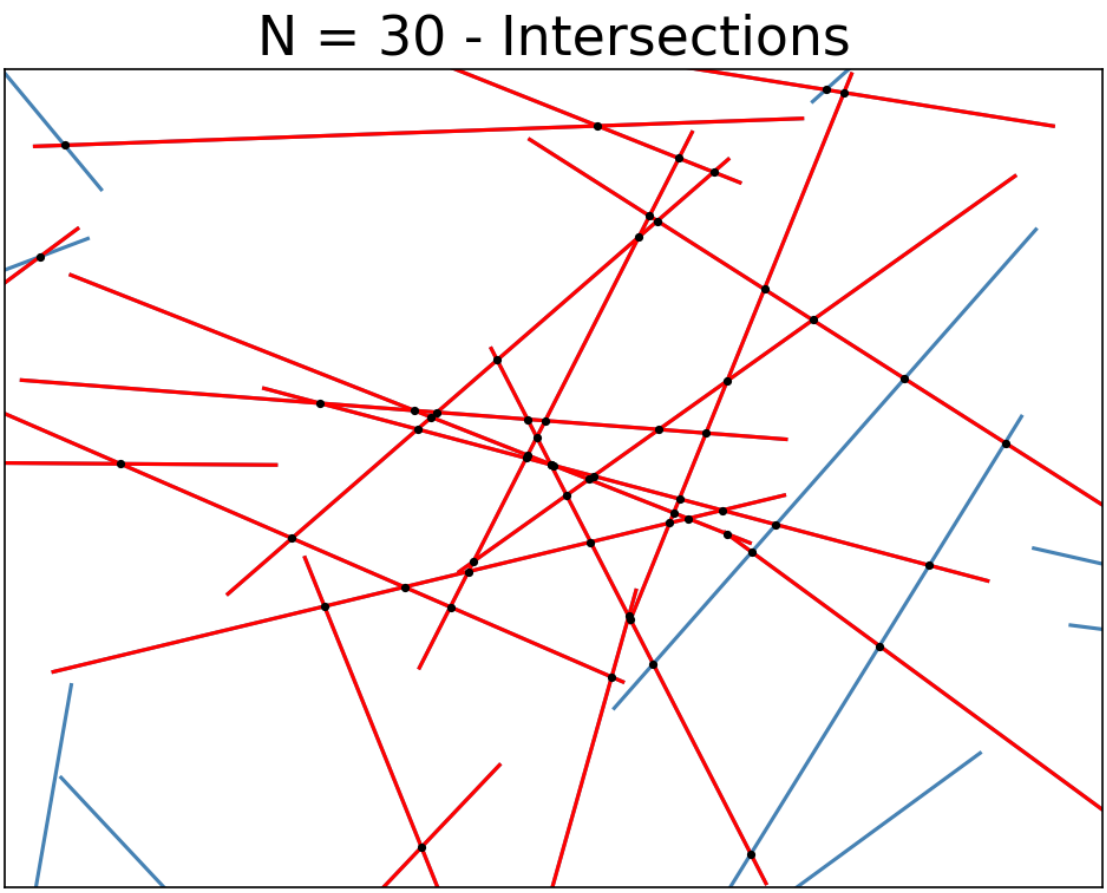


Fig. — 02 intersections N30

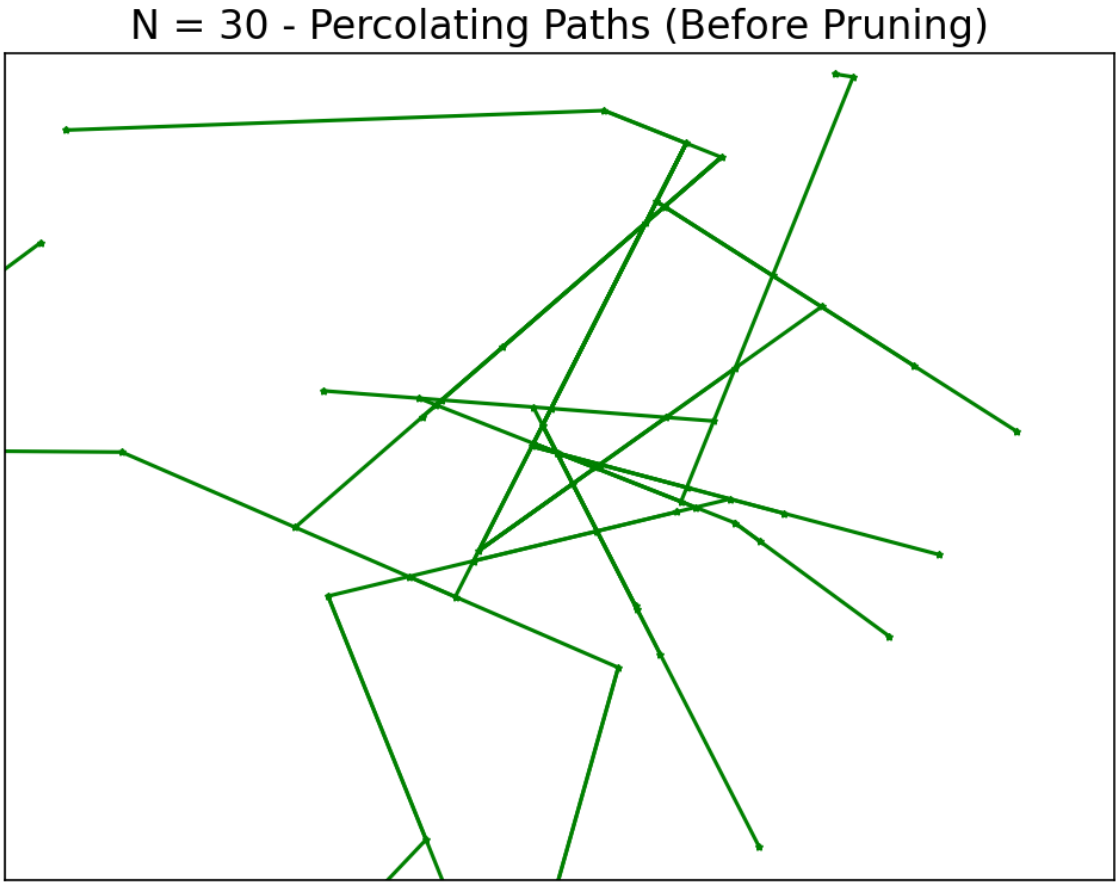


Fig. — 03 percolating paths before pruning N30

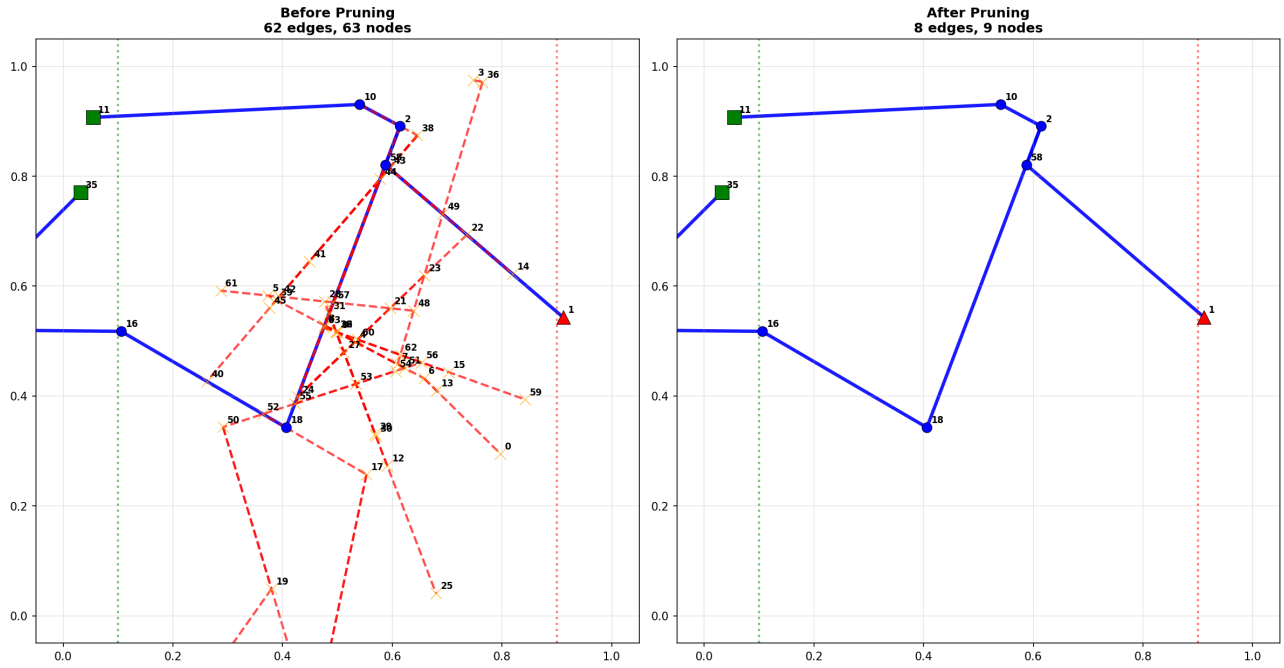


Fig. — 04 before after pruning N30

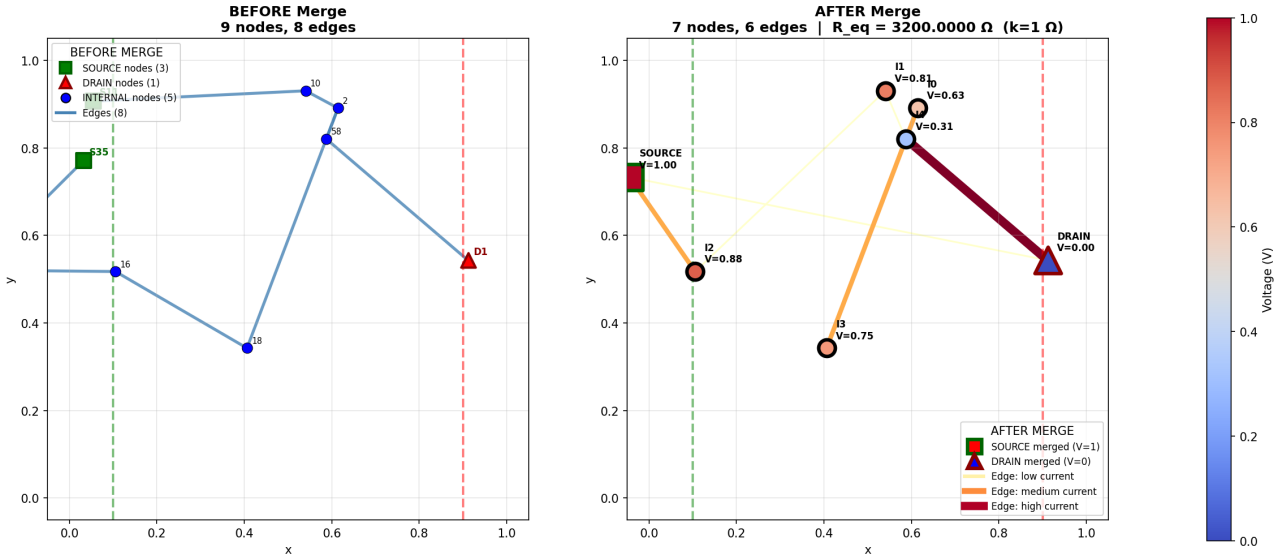


Fig. — 05 before after merge N30

3. Edge Resistances (Pruned Network)

Edge #	Node 1	Node 2	Conductance (S)	Resistance (Ω)	Voltage Drop (V)
0	11	10	1.0000	1.0000	0.000000
1	10	2	1.0000	1.0000	0.875000
2	2	18	1.0000	1.0000	0.875000
3	2	58	1.0000	1.0000	0.875000
4	18	16	1.0000	1.0000	0.000000
5	58	1	1.0000	1.0000	0.812500
6	16	37	1.0000	1.0000	0.000000
7	37	35	1.0000	1.0000	0.000000

4. Impedance Calculation

Quantity	Value
Applied voltage (V)	1.00 V
Resistance per segment k	1 Ω
Calculated impedance R_eq	3200.000000 Ω
Total current I	-1.250000e-04 A

Impedance for different k (resistance per segment) values

k (Ω/segment)	R_equivalent (Ω)
1	3.200000
10	32.000000
100	320.000000

k (Ω/segment)	R_equivalent (Ω)
1000	3200.000000
10000	32000.000000